

1. ERROR HANDLING

No file names passed

```
ashithadmin@LAPTOP-OAE19R6D:~/Projects/Inverted_Search$ ./a.out
```

```
Error: Invalid argument : ./a.out
```

```
Usage:
```

```
Valid Input      : ./a.out <.txt file-1> <.txt file-2> ...
```

```
File requirements : Entered file should exist, non-empty and non-duplicate.
```

No valid file names passed

```
ashithadmin@LAPTOP-OAE19R6D:~/Projects/Inverted_Search$ ./a.out f1 f2 f3
```

```
Invalid / duplicate / empty / non-existent files => "f1" "f2" "f3"
```

```
Error: Entered files are not valid.
```

```
Usage:
```

```
Valid Input      : ./a.out <.txt file-1> <.txt file-2> ...
```

```
File requirements : Entered file should exist, non-empty and non-duplicate.
```

2. CREATE DATABASE

Database can only be created once

```
ashithadmin@LAPTOP-OAE19R6D:~/Projects/Inverted_Search$ ./a.out f1.txt f2.txt f3.txt f12 f4.c f5.txt
Invalid / duplicate / empty / non-existent files => "f12" "f4.c" "f5.txt"
Valid files => "f1.txt" "f2.txt" "f3.txt"

::::::::::::INVERTED SEARCH MENU::::::::::::

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

::::::::::::

Enter Your Choice: 1

Created Database Successfully.
```

```
::::::::::::INVERTED SEARCH MENU::::::::::::

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

::::::::::::

Enter Your Choice: 1

Already Database is Created.
```

3. DISPLAY DATABASE

Display database after creating

::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

::

Enter Your Choice: 2

Index		Word	File Count	File Name[Word Count]
Word 1:	0	are	2	f1.txt[1] f2.txt[1]
Word 2:	7	hi	3	f1.txt[2] f2.txt[1] f3.txt[1]
Word 3:	7	Hello	1	f1.txt[1]
Word 4:	7	how	3	f1.txt[1] f2.txt[1] f3.txt[1]
Word 5:	7	hello	1	f3.txt[1]
Word 6:	22	what?	1	f3.txt[1]
Word 7:	24	you	1	f2.txt[1]
Word 8:	26	?	1	f2.txt[1]
Word 9:	26	12345	1	f3.txt[1]

4. SEARCH DATABASE

Searching word in database:
If found and if not found

```
::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::
```

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

```
::::::::::::::::::::
```

Enter Your Choice: 3
Enter the word to search: how

Index	Word	File Count	File Name[Word Count]
7	how	3	f1.txt[1] f2.txt[1] f3.txt[1]

Word -> "how" is found in database.

```
::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::
```

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

```
::::::::::::::::::::
```

Enter Your Choice: 3
Enter the word to search: hai

Word -> "hai" is not found in database.

5. SAVE DATABASE

Invalid File name and if valid with non-empty file, then asks for overwriting permission

```
::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::
1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

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Enter Your Choice: 4
Enter the file name to store database: backup

Error: Entered file backup do not extension.

::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::
1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

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Enter Your Choice: 4
Enter the file name to store database: backup.txt

Do you want to overwrite backup.txt file (Yes - 1 / No - 0): 0

Overwriting cancelled.
```

Overwriting file contents with current database data

```
::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::  
  
1. Create Database.  
2. Display Database.  
3. Search Database.  
4. Save Database.  
5. Update Database.  
6. Exit.  
  
::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::  
  
Enter Your Choice: 4  
Enter the file name to store database: backup.txt  
  
Do you want to overwrite backup.txt file (Yes - 1 / No - 0): 1  
  
Overwriting the "backup.txt" file content.  
  
Saved to Database Successfully.
```


After saving database data to backup.txt file

```
ashithadmin@LAPTOP-OAE19R6D:~/Projects/Inverted_Search$ ls
Create_Database.c  Inverted_search.h  Save_Database.c  Update_database.c  a.out      f1.txt  f3.txt  main.c
Display.c          Makefile           Search.c          Validate_args.c    backup.txt  f2.txt  f6.txt
ashithadmin@LAPTOP-OAE19R6D:~/Projects/Inverted_Search$ cat backup.txt
#0;are;3;f1.txt;1;f2.txt;1;f6.txt;1;#
#7;hi;3;f1.txt;2;f2.txt;1;f3.txt;1;#
#7;Hello;1;f1.txt;1;#
#7;how;4;f1.txt;1;f2.txt;1;f3.txt;1;f6.txt;1;#
#7;hello;2;f3.txt;1;f6.txt;1;#
#7;hai;1;f6.txt;1;#
#20;u?;1;f6.txt;1;#
#22;what?;1;f3.txt;1;#
#24;you;1;f2.txt;1;#
#26;?;1;f2.txt;1;#
#26;12345;1;f3.txt;1;#
ashithadmin@LAPTOP-OAE19R6D:~/Projects/Inverted_Search$ |
```

6. UPDATE DATABASE

After saving database data to backup.txt file

```
::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::
```

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

```
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```

Enter Your Choice: 5

Enter file name to update database: f6.txt

Error: After creating database update database not possible

Trying to update database after database already created

```
ashithadmin@LAPTOP-OAE19R6D:~/Projects/Inverted_Search$ ./a.out f1.txt f2.txt f3.txt f12 f4.c f5.txt f6.txt
```

```
Invalid / duplicate / empty / non-existent files => "f12" "f4.c" "f5.txt"
```

```
Valid files => "f1.txt" "f2.txt" "f3.txt" "f6.txt"
```

```
::::::::::::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::::::::::
```

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

```
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```

```
Enter Your Choice: 5
```

```
Enter file name to update database: f6.txt
```

```
Error: Updating wrong backup file.
```

Invalid file passed to update database

```
:::::::::::::::::::::INVERTED SEARCH MENU:::::::::::::::::::::

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

:::::::::::::::::::::

Enter Your Choice: 5
Enter file name to update database: backup.txt

Database updated successfully.

:::::::::::::::::::::INVERTED SEARCH MENU:::::::::::::::::::::

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

:::::::::::::::::::::

Enter Your Choice: 1

Already updated files => "f1.txt" "f2.txt" "f3.txt"

Database successfully created files => "f6.txt"
```

Displaying database after updating and then creating of database

::::::::::::INVERTED SEARCH MENU::::::::::::

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

::::::::::::

Enter Your Choice: 2

Index	Word	File Count	File Name[Word Count]
Word 1: 0	are	3	f1.txt[1] f2.txt[1] f6.txt[1]
Word 2: 7	hi	3	f1.txt[2] f2.txt[1] f3.txt[1]
Word 3: 7	Hello	1	f1.txt[1]
Word 4: 7	how	4	f1.txt[1] f2.txt[1] f3.txt[1] f6.txt[1]
Word 5: 7	hello	2	f3.txt[1] f6.txt[1]
Word 6: 7	hai	1	f6.txt[1]
Word 7: 20	u?	1	f6.txt[1]
Word 8: 22	what?	1	f3.txt[1]
Word 9: 24	you	1	f2.txt[1]
Word 10: 26	?	1	f2.txt[1]
Word 11: 26	12345	1	f3.txt[1]

7. EXIT MENU

Exiting the Inverted Search Menu

```
::::::::::::::::::::INVERTED SEARCH MENU::::::::::::::::::::
```

1. Create Database.
2. Display Database.
3. Search Database.
4. Save Database.
5. Update Database.
6. Exit.

```
::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::
```

Enter Your Choice: 6

Exiting...